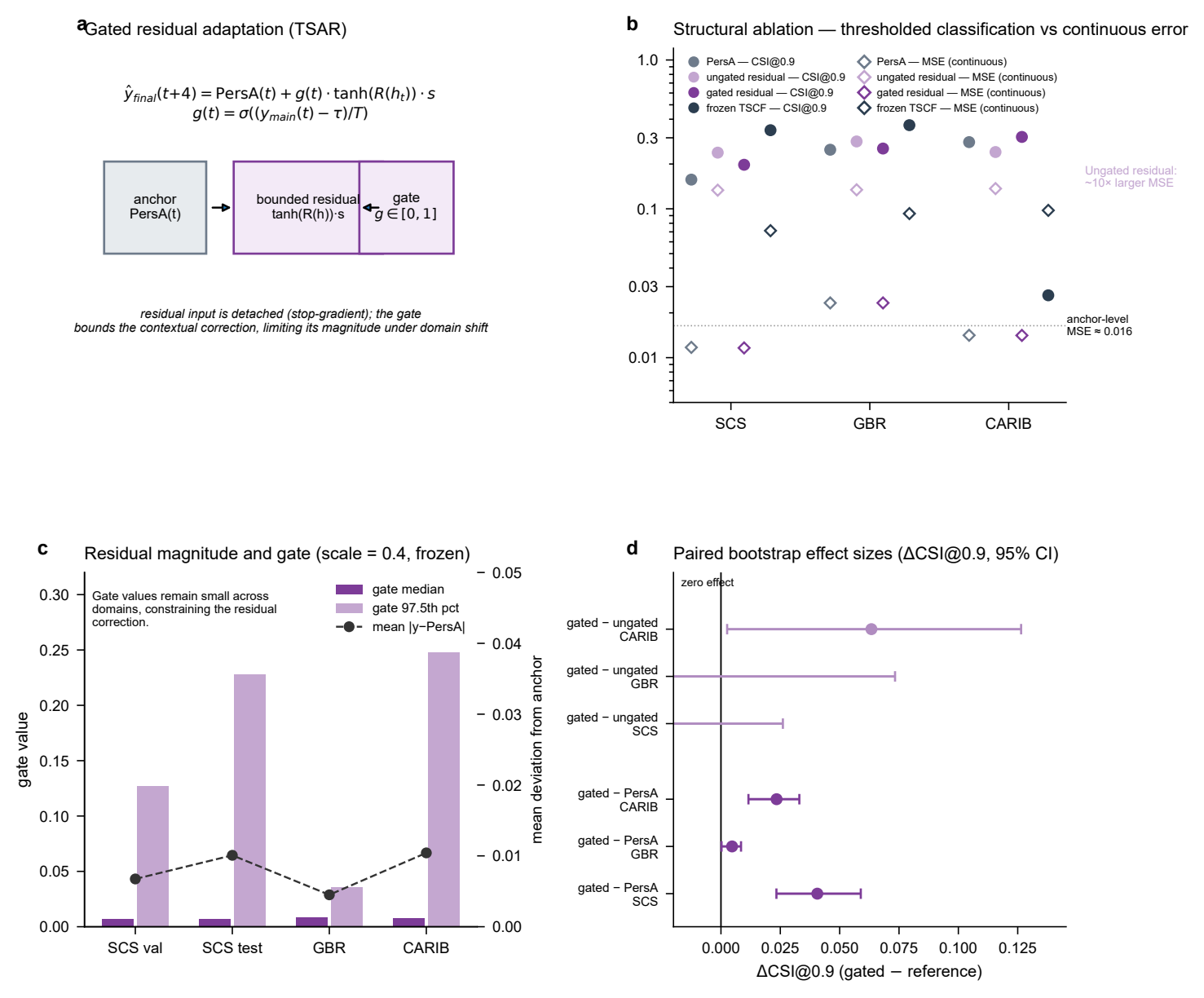

a
Gated residual adaptation (TSAR)
$\hat{y}_{final}(t+4) = \text{PersA}(t) + g(t) \cdot \tanh(R(h_t)) \cdot s$
$g(t) = \sigma((y_{main}(t) - \tau)/T)$
anchor
PersA(t)
bounded residual
tanh(R(h))·s
gate
$g \in [0, 1]$
residual input is detached (stop-gradient); the gate
bounds the contextual correction, limiting its magnitude under domain shift
b
Structural ablation — thresholded classification vs continuous error
PersA — CSI@0.9
ungated residual — CSI@0.9
gated residual — CSI@0.9
frozen TSCF — CSI@0.9
PersA — MSE (continuous)
ungated residual — MSE (continuous)
gated residual — MSE (continuous)
frozen TSCF — MSE (continuous)
1.0
0.3
0.1
0.03
0.01
Ungated residual:
~10× larger MSE
anchor-level
MSE ≈ 0.016
SCS
GBR
CARIB
c
Residual magnitude and gate (scale = 0.4, frozen)
Gate values remain small across
domains, constraining the residual
correction.
gate median
gate 97.5th pct
mean |y−PersA|
gate value
mean deviation from anchor
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.05
0.04
0.03
0.02
0.01
0.00
SCS val
SCS test
GBR
CARIB
d
Paired bootstrap effect sizes (ΔCSI@0.9, 95% CI)
zero effect
gated − ungated
CARIB
gated − ungated
GBR
gated − ungated
SCS
gated − PersA
CARIB
gated − PersA
GBR
gated − PersA
SCS
0.000
0.025
0.050
0.075
0.100
0.125
ΔCSI@0.9 (gated − reference)